

CSE422

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Sec - 09

Answer to the question no:1 .

$C=6$

	1	2	3	4	5
1					
2		Q_2		Q_4	
3	Q_1				
4		Q_3			
5				Q_5	

	Q_2			
			Q_4	
Q_1				
		Q_3		
				Q_5

goal

			Q_1	
	Q_2		Q_4	
Q_1				
	Q_3			
		Q_5		

move left

	Q_2		Q_4	
Q_1				
	Q_3	Q_5		

move up

It will not move any further and get stuck in local maximer.

Answer to the question no: 2.

① In the graph, the x node is not connected to any other nodes. So, heuristic value cannot ~~detect~~ detect or ~~as~~ estimate to an detached or isolated one. Because they don't needed in any of the searching algorithm. So, we can't determine/estimate a heuristic value of node x .

② By observing heuristic value of $h_1(n)$ and $h_2(n)$, ~~no~~ $h_2(n)$ is the dominant heuristic. $h_2(n) \geq h_1(n)$. So, $h_2(n)$ is better for A^* search.

$$\boxed{A^5}$$

$$A^5 \quad \boxed{B^4 \quad C^7 \quad D^7}$$

$$B^4 \quad \boxed{C^7 \quad D^7 \quad D^5 \quad E^7}$$

$$D^5 \quad \boxed{C^7 \quad D^7 \quad E^7 \quad E^5 \quad F^9 \quad G^6}$$

$$E^5 \quad \boxed{C^7 \quad D^7 \quad E^7 \quad F^9 \quad G^6 \quad G^5}$$

$$G^6 \quad \boxed{C^7 \quad D^7 \quad E^7 \quad F^9 \quad G^6}$$

$$B = 3 + 1 = 4$$

$$C = 5 + 2 = 7$$

$$D = 2 + 5 = 7$$

$$D = 2 + 2 + 1 = 5$$

$$E = 1 + 5 + 1 = 7$$

$$E = 1 + 1 + 2 + 1 = 5$$

$$F = 3 + 3 + 2 + 1 = 9$$

$$G = 0 + 3 + 1 + 2 = 6$$

$$G = 0 + 1 + 2 + 1 + 1 = 5$$

$$A \xrightarrow{1} B \xrightarrow{2} D \xrightarrow{1} E \xrightarrow{1} G$$

$$\text{Path cost} = 1 + 2 + 1 + 1 = 5$$

② The possible path from node c to node E is:

① $C \rightarrow D \rightarrow E$

② $C \rightarrow F \rightarrow D \rightarrow E$

③ $C \rightarrow F \rightarrow G \rightarrow E$

④ $C \rightarrow D \rightarrow G \rightarrow E$

The determine path can be optimal if all the heuristic value is ~~admi~~ admissible.

$$h(n) \leq \text{actual distance}$$

for ① the actual distance $D_1 = 3 + 1 = 4$

for ② " " " $D_2 = 4 + 3 + 1 = 8$

for ③ " " " $D_3 = 4 + 3 + 1 = 8$

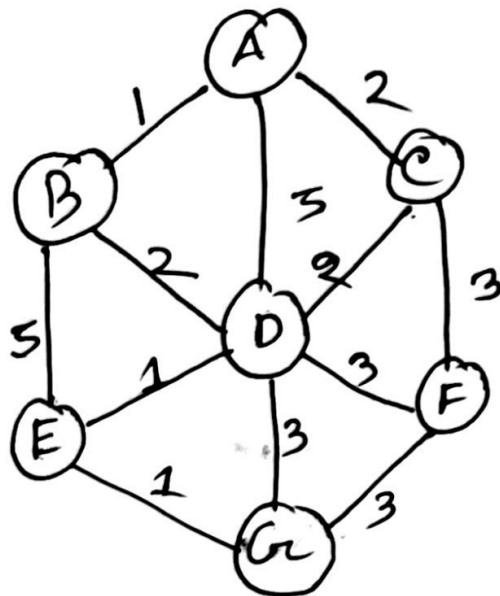
for ④ " " " $D_4 = 3 + 3 + 1 = 7$

$$h_1(c) = 4 \quad h_2(c) = 5$$

$$h_1(c) \leq D_1 \quad \text{and} \quad h_2(c) \leq D_2, D_3, D_4$$

So, the optimal path or sub optimal path can be possible to determine. Because the heuristic value are close ~~and going to~~ and proved the condition.

Q1



State	$h_1(n)$	$h_2(n)$
A	5	5
B	2	3
C	4	5
D	2	2
E	1	1
F	2	3
G	0	0

We know,

$$h(B) \leq \text{weight}(B, D) + h(D)$$

and

$$h(B) \leq \text{weight}(B, E) + h(E)$$

~~$h(B)$~~ $h_1(B)$

$$2 \leq (2+2)$$

and

$$2 \leq (5+1)$$

$h_2(B)$

$$3 \leq (2+2)$$

and.

$$3 \leq (5+1)$$

$\therefore h_2(B)$ value is consistent even the cost from C to F and C to D is changed.